

Explainable AI for NBA Win Prediction

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Abstract. Artificial intelligence (AI) and machine learning (ML) have enabled increasingly sophisticated sports prediction models, yet most prior work treats outcome prediction as either a pre-game forecasting task or an in-game updating problem. This separation limits practical use for fans and analysts who want to identify games that are likely to remain competitive and to understand how that assessment evolves as new evidence emerges during play. In this paper, we present an evidence-to-system framework that bridges pre-game and in-game prediction with an emphasis on interpretability and human-centered use. We first conduct a PRISMA-guided systematic review (2006–2025) of AI-based basketball outcome prediction, focusing on modeling approaches, temporal feature integration, and interpretability strategies. Motivated by review findings, we then develop a proof-of-concept NBA win-prediction dashboard that initializes pre-game win probabilities using an RBF-kernel Support Vector Machine (SVM) trained on season statistics, rolling recent-form features, and pre-game Elo ratings. For transparency, we compute SHAP-based local feature attributions and generate short natural-language rationales for each prediction. In evaluation of the pre-game pipeline, the SVM achieves 79%–81% validation accuracy across two settings, supporting its role as a stable baseline prior for adaptive systems. Overall, our results highlight phase-dependent model behavior and the importance of explanations that translate probabilistic outputs into basketball-relevant factors, contributing toward more practical and interpretable sports analytics.

Keywords: Sports analytics · Explainable AI · Pre-game and Real-time prediction · Human-centered AI · NBA outcome prediction

1 Introduction

Artificial intelligence (AI) and machine learning (ML) have transformed how athletes, coaches, analysts, and fans interpret performance in modern sports. In professional basketball, and particularly in the NBA, the availability of detailed player- and team-level data has enabled increasingly sophisticated predictive models capable of estimating game outcomes, tracking performance trends, and

summarizing complex game dynamics. Despite this progress, much of the existing work treats prediction as either a pre-game forecasting task or an in-game updating problem, with relatively few systems that integrate both perspectives within a single adaptive and interpretable framework.

From the perspective of sports fans and analysts, this fragmentation limits practical usefulness. Many games appear competitive before tip-off but quickly become one-sided, while others remain uncertain until the final minutes. As students and fans with limited time to follow every game in full, we are motivated by a simple but underexplored question: which games are likely to remain close, and how does that assessment evolve as the game unfolds? Addressing this question requires models that not only provide an initial assessment of competitiveness before a game begins, but also adapt in real time as new evidence accumulates. Equally important, these predictions must remain understandable to human users so that changes in outcome likelihood can be interpreted in terms of meaningful basketball events rather than opaque numerical shifts.

This study examines how predictive models can evolve in real time using live game data while remaining interpretable and transparent to users. Rather than focusing solely on predictive accuracy, we emphasize adaptability, efficiency, and explainability, recognizing that trust and effective decision-making depend on understanding why a model’s prediction changes. We compare multiple machine learning approaches—Support Vector Machines (SVM), Random Forest, XGBoost, and LightGBM—evaluating not only their accuracy but also their suitability for different stages of the game, from pre-game context to live, momentum-driven updates.

Grounded in a PRISMA-guided systematic review and supported by a proof-of-concept NBA win-prediction dashboard, this work seeks to identify modeling strategies that balance predictive performance with human-centered interpretability. While the immediate application is NBA basketball, the broader goal is to develop a generalizable framework that can be extended to other sports with similar temporal structure, enabling adaptive and explainable outcome prediction across domains.

Guided by this motivation, the study addresses three research questions that align with the system design, evaluation, and synthesis presented in this paper:

- **Algorithmic Adaptability:** Which machine learning algorithm—among SVM, KNN, Random Forest, XGBoost, and LightGBM—best adapts to both pre-game and in-game NBA win prediction when using season averages, momentum-based statistics, and live data?
- **Feature Integration:** To what extent does incorporating in-game performance metrics improve predictive accuracy and adaptability compared to models based solely on static season averages?
- **Explainability and Human Trust:** How can interpretability techniques, such as SHAP or feature-importance analysis, enhance user understanding and trust in AI-driven predictions, transforming model outputs from black-box estimates into transparent decision-support tools?

Together, these questions aim to identify prediction strategies that are not only accurate, but also adaptive and explainable, supporting meaningful human–AI collaboration in sports analytics. By integrating pre-game context, live updates, and human-centered explanation within a single framework, this work contributes toward more practical, interpretable, and engaging AI-assisted sports analysis.

Table of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
ML	Machine Learning
NBA	National Basketball Association
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
SVM	Support Vector Machine
KNN	k-Nearest Neighbors
RF	Random Forest
GNN	Graph Neural Network
GCN	Graph Convolutional Network
GAT	Graph Attention Network
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
SHAP	SHapley Additive exPlanations
XAI	Explainable Artificial Intelligence
HCI	Human–Computer Interaction
HCII	Human–Computer Interaction International
Elo (ELO)	Elo Rating System
FG%	Field Goal Percentage
3P%	Three-Point Percentage
FT%	Free Throw Percentage
PTS	Points Scored
REB	Rebounds
AST	Assists
TOV	Turnovers
API	Application Programming Interface
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
RBF	Radial Basis Function
KPI	Key Performance Indicator

2 Related Works

Research in sports analytics spans classical machine learning, ensemble and graph-based modeling, and increasingly real-time prediction with explainability. In basketball outcome prediction, traditional supervised methods such as

logistic regression and Support Vector Machines (SVM) remain common baselines due to their simplicity, stability, and reproducibility [7][8][11]. More recent studies demonstrate that predictive performance is strongly influenced by feature design, with efficiency-based indicators and box-score features such as shooting efficiency and rebounding repeatedly emerging as high-impact predictors [4][5][6]. Reviews of team-sport prediction further highlight that tree-based and ensemble approaches frequently outperform simpler baselines, while also noting persistent limitations in interpretability and real-time adaptability [12][9] [29]. Building on this trend, He and Choi [14] demonstrate the value of stacked ensemble architectures for NBA outcomes, and Alves and Barbosa [13] provide modern benchmarking across NBA and WNBA contexts, illustrating that achievable accuracy depends heavily on league conditions and feature quality.

Several works argue that outcome prediction benefits from representing league structure rather than treating matchups as independent samples. Zhao et al. [1] fuse Graph Convolutional Networks with Random Forests to capture inter-team relationships in multi-season NBA data, demonstrating performance gains over flat tabular baselines. This direction is extended by Zhao et al. [15], who incorporate Graph Attention Networks with clustering-based feature selection, reinforcing that relational structure and reduced feature redundancy can improve predictive stability.

Live and stage-aware prediction has received growing attention, motivated by the fact that game context evolves rapidly; similar real-time win prediction has also been demonstrated in adjacent domains such as esports [3]. Recent work also frames win prediction as a streaming task using sliding windows and explainable updates, aligning closely with real-time deployment needs [23]. Ouyang et al. [2] integrate XGBoost with SHAP for real-time NBA forecasting and show that feature relevance shifts by game phase: assist-related indicators tend to dominate early, while offensive rebounding and three-point efficiency become more salient later. These findings align with broader evidence that basketball dynamics are nonstationary and context-dependent [17], suggesting that static models may fail to capture evolving in-game drivers. Despite this progress, much prior work addresses pre-game prediction, live updating, and interpretability as separate problems, with limited research that unifies phase-aware prediction and human-centered explanation within a single end-to-end system.

Methodologically, several foundational approaches model games as sequential, state-dependent processes. Logistic regression and Markov chain formulations enable principled win-probability estimation from play-by-play structure and possessions [19] [20] [18]. Win-probability frameworks have also been applied to evaluate late-game decision-making, highlighting how outcome likelihood can shift sharply in end-of-game scenarios [16]. At a finer granularity, Sano et al. [22] show that tactical micro-events such as pass receivability can be predicted from player and ball motion, supporting the feasibility of real-time predictive overlays beyond final outcomes. Complementing these approaches, spatiotemporal tracking data provides additional live signal; Jiao et al. [21] demonstrate that

Table 1. Summary of representative studies relevant to AI-based sports outcome prediction and interpretability. The table highlights differences in modeling approach, interpretability strategy, data type, and human–AI emphasis across domains.

Study	Sport / Domain	Common AI Models	Mod- Interpretability Methods	Human–AI Focus
Hammes et al. (2022)	Multi-sport (review)	Random Forest, XGBoost, DNN	Conceptual work, narrative analysis	frame- Trust, usability, and deployment challenges
Lalwani et al. (2022)	Volleyball	XGBoost, rule-based models	SHAP explanations	Model transparency for analysts
Pitassi et al. (2024)	Ice Hockey (NHL)	Explainable ensembles, regression models	Stakeholder-centered explanation design	Human–AI collaboration and visualization
Ouyang et al. (2024)	NBA Basketball	XGBoost	SHAP (local explanations)	Phase-aware explanation in live prediction
Zhao et al. (2023, NBA Basketball 2024)		GCN, GAT, Random Forest	Feature importance analysis	Structural modeling and interpretability
This work	NBA Basketball	SVM (pre-game), XGBoost (in-game)	SHAP-based local explanations	Human-centered, phase-aware decision support

SportVU-derived trajectories encode possession-level patterns beyond standard box-score features.

As predictive systems move toward deployment, explainability and human-centered design become central concerns. Du and Yuan [24] survey sports visualization and visual analytics, emphasizing that interpretability and interaction support sense-making in complex performance data. More broadly, human-centered XAI research shows that explanation quality must be evaluated with users, and that transparency alone does not guarantee calibrated trust [25]. Related HCII findings further suggest that trust and understanding can diverge depending on explanation design and user biases. Practitioner-facing explainability research argues that explanations aligned with domain concepts can improve acceptance even when accuracy gains are modest [28]. In sports contexts, reviews identify interpretability, data quality, and workflow integration as persistent barriers to adoption despite advances in predictive accuracy [27][26]. Finally, the PRISMA 2020 guidelines provide updated methodological standards that strengthen transparency and reproducibility when synthesizing evidence across diverse studies [10].

Together, this literature demonstrates substantial progress in both pre-game and in-game modeling, yet these contributions are often developed in isolation and are not consistently paired with user-centered explanations. Our work builds on these findings by integrating pre-game and in-game prediction within a single framework and pairing phase-aware forecasting with explainable outputs designed to support transparent, human-centered interpretation. We further include SVM-based modeling as a computationally efficient and widely adopted baseline, enabling clear comparison while supporting live, stage-aware updates.

3 Methods

3.1 Systematic Review Framework

This study adopts the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to ensure transparency, reproducibility, and methodological rigor in synthesizing prior research on AI-based basketball outcome prediction. PRISMA was selected due to its suitability for systematically organizing evidence across interdisciplinary domains, including machine learning, sports analytics, visualization, and human–AI interaction. Given the methodological diversity of this literature—spanning static pre-game models, live in-game forecasting, and explainability techniques—PRISMA provides a standardized process for documenting study identification, screening, eligibility assessment, and inclusion, while minimizing selection bias and improving replicability.

The review protocol followed the four PRISMA phases—identification, screening, eligibility, and inclusion—with all decisions documented to enable traceability from the initial corpus to the final set of analyzed studies.

3.2 Literature Search Strategy

A structured literature search was conducted primarily using Google Scholar, which offers broad coverage of peer-reviewed journals and full-length conference proceedings commonly cited in sports analytics and applied machine learning research. This includes venues such as *PLOS ONE*, MDPI journals, ACM conference proceedings, IEEE Transactions, Springer Nature, and Elsevier publications. The choice of Google Scholar aligns with established practice in recent systematic reviews of sports analytics and explainable AI, where relevant work is distributed across heterogeneous publication outlets.

The search was conducted in October 2025 and included studies published between 2006 and 2025, capturing both foundational work on win-probability modeling and contemporary research on live prediction and explainable AI.

Search queries were designed around three core concept clusters aligned with the study’s research questions:

- **Machine Learning and Explainable AI:** machine learning, support vector machine, ensemble learning, random forest, XGBoost, LightGBM, graph neural networks, explainable AI, SHAP, feature importance.
- **Basketball Analytics and Outcome Prediction:** NBA prediction, basketball outcome prediction, win probability, sports forecasting, play-by-play modeling, in-game prediction.
- **Interpretability and Human–AI Interaction:** model transparency, interpretability, user trust, human-centered AI, explainable dashboards, sports visualization.

These keyword groups were combined using Boolean operators to identify studies at the intersection of predictive modeling, live or stage-aware forecasting,

and interpretability. Backward and forward citation tracking was additionally employed on high-impact papers to identify relevant studies not captured by the initial queries.

3.3 Inclusion and Exclusion Criteria

Studies were screened using predefined inclusion and exclusion criteria to ensure both relevance and methodological quality. A study was included if it satisfied all of the following conditions:

- Applied machine learning or AI methods (e.g., SVM, ensemble models, graph-based models, hybrid approaches) to sports or basketball outcome prediction.
- Addressed temporal adaptability, including live, play-by-play, stage-aware, or momentum-based features, or explicitly discussed limitations of static season-average modeling.
- Implemented or analyzed model interpretability, including feature importance methods, SHAP-based explanations, rule-based reasoning, or human-centered explanation strategies.
- Was published as a peer-reviewed journal article or full-length conference paper.
- Reported quantitative evaluation metrics (e.g., accuracy, AUC, F1-score) using a clearly described dataset and evaluation protocol.

Studies were excluded if they relied exclusively on traditional statistical heuristics without ML/AI components, lacked methodological transparency, omitted reproducible performance results, were short abstracts or posters without full methodological detail, or were non-English publications without accessible full text. The complete inclusion and exclusion criteria are summarized in Table 2.

3.4 Study Selection and Categorization

Following duplicate removal, 30 studies were retained for full-text screening and inclusion. Each study was independently reviewed and categorized along two primary analytical dimensions:

Modeling Approach. Studies were grouped according to their predictive modeling strategy, including classical supervised models (e.g., SVM, logistic regression), tree-based and ensemble methods (e.g., Random Forest, XGBoost, LightGBM), and structured or relational models (e.g., graph neural networks, possession-based Markov models).

Interpretability Strategy. Studies were categorized based on how model transparency was addressed, including global feature importance analysis, local explanation methods such as SHAP, and explicit discussion of explanation design, usability, or human-centered transparency.

This categorization enabled systematic comparison across both predictive performance and explanatory capacity, directly supporting analysis aligned with the study’s three research questions.

Table 2. Inclusion and exclusion criteria applied during the PRISMA screening process for studies on AI-based sports outcome prediction.

Inclusion / Exclusion Criteria	Include?
Applies artificial intelligence (machine learning, ensemble, or hybrid models) to sports or basketball outcome prediction	Yes
Incorporates live or momentum-based performance features beyond static season averages	Yes
Implements or discusses model interpretability (e.g., SHAP, feature importance, rule-based explanation)	Yes
Published in a peer-reviewed journal or full-length conference paper	Yes
Includes quantitative evaluation metrics (e.g., accuracy, AUC, F1-score) and clear dataset description	Yes
Written in English with accessible full text	Yes
Focuses solely on traditional statistical or heuristic prediction without ML/AI techniques	No
Lacks methodological transparency or reproducible performance results	No

3.5 Synthesis Approach

A mixed-method synthesis strategy was employed, combining quantitative comparison with qualitative thematic analysis. Quantitatively, reported performance metrics—such as prediction accuracy, AUC, and robustness across game phases—were compared to identify broad performance trends rather than absolute algorithm rankings. This approach acknowledges heterogeneity in datasets, seasons, feature sets, and evaluation protocols, which precludes direct metric aggregation.

Qualitatively, studies were coded for recurring themes related to live-data integration, adaptability, transparency, usability, and trust. Coding was informed by established human-centered AI literature, with particular attention to how explanations were framed, evaluated, and contextualized for end users. Rather than conducting a formal meta-analysis, which is impractical given methodological diversity, the synthesis focuses on identifying design patterns, methodological gaps, and implications for real-time, explainable sports analytics systems.

3.6 System Design Informed by the Review

Findings from the systematic review directly informed the design of a proof-of-concept NBA win-prediction dashboard. The system integrates pre-game season-level statistics with in-game, play-by-play-derived features to update win probabilities in real time. Explainable AI techniques are employed to surface key

Table 3. Keyword groupings used in the PRISMA literature search for finding sources.

Category	Representative Keywords
Machine Learning / XAI	machine learning, support vector machine, random forest, XG-Boost, LightGBM, graph neural networks, explainable AI, SHAP, feature importance
Basketball Analytics	NBA prediction, basketball outcome prediction, win probability, sports forecasting, play-by-play modeling, in-game prediction
Interpretability / HCI	model transparency, interpretability, user trust, human-centered AI, explainable dashboards, sports visualization

contributing factors underlying each prediction, enabling users to observe how feature importance shifts as the game progresses.

This system operationalizes insights from the reviewed literature by demonstrating how adaptive prediction and interpretability can be combined within a single human-centered interface, bridging empirical evidence from prior studies with a practical, interactive application. A PRISMA flow diagram summarizing the study selection process is presented in Figure 1, and the keyword groupings used during screening are reported in Table 3, providing full transparency into the review methodology and its role in shaping both analytical focus and system design decisions.

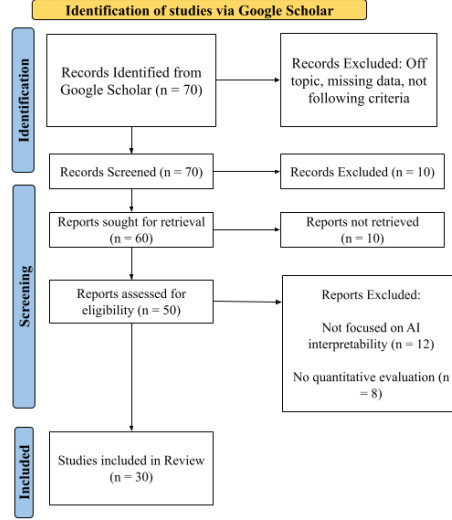


Fig. 1. PRISMA 2020 flow diagram summarizing the identification, screening, eligibility, and inclusion of studies in the systematic review.

4 Results

4.1 PRISMA Review Outcomes (R0)

Using the PRISMA 2020 screening and categorization process described in the Methods section, the systematic literature synthesis revealed three consistent patterns relevant to NBA outcome prediction. First, tree-based ensemble methods—particularly Random Forest, XGBoost, and LightGBM—frequently report strong predictive performance when feature engineering is well designed and evaluation is conducted using realistic season-based splits. Second, the inclusion of “recent form” or time-windowed performance features often improves predictive quality relative to models relying solely on static season averages. Third, while explainability techniques are increasingly incorporated, many studies provide limited user-facing justification for why a prediction is produced, especially in basketball-specific and real-time prediction pipelines.

These findings informed both the selection of modeling approaches and the design priorities for interpretability in the implemented system.

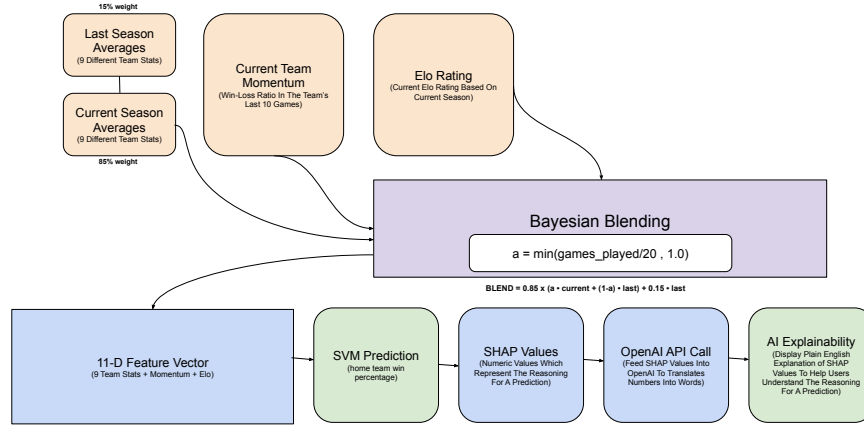


Fig. 2. Pre-game prediction pipeline overview. Team statistics, momentum (rolling-window recent form), and pre-game Elo are transformed into a home-away feature difference vector and used as input to an RBF-kernel SVM. SHAP feature attributions are computed for each prediction and converted into natural-language rationales via the OpenAI API.

4.2 RQ1: Algorithm Selection for Pre-Game Prediction

To identify a suitable pre-game prediction model, we evaluated multiple supervised learning approaches commonly reported in the reviewed literature, including Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest, XGBoost, and LightGBM. Model comparison emphasized not only predictive accuracy but also stability, reproducibility, and suitability for deployment as a pre-game baseline.

Across two evaluation settings, the RBF-kernel SVM demonstrated the most consistent performance:

- 79% validation accuracy in an evaluation based on the 2024–2025 NBA season.
- 81% validation accuracy in the current pre-game prediction pipeline.

These results align with trends identified in the PRISMA synthesis, where SVMs frequently serve as competitive and stable baselines for pre-game outcome prediction. Based on this evidence, the SVM was selected as the pre-game model for the system.

4.3 RQ2: Feature Set for the Pre-Game Model

The selected SVM operates on a pre-game feature vector constructed as the difference between home-team and away-team statistics. This representation em-

phasizes relative team strength while maintaining a consistent feature space across matchups.

The feature set includes:

- **Team performance statistics:** field goal percentage (FG%), three-point percentage (3P%), free throw percentage (FT%), points scored (PTS), rebounds (REB), assists (AST), turnovers (TOV), plus/minus, and net margin.
- **Momentum and recent form:** rolling-window performance indicators computed over the previous ten games, including recent win percentage.
- **Team strength rating:** a pre-game Elo score (ELO_PRE) computed from chronological game outcomes and included as a summary signal of evolving team quality.

This feature design reflects findings from the literature review, which consistently identifies efficiency metrics, rebounding, and recent performance trends as high-impact predictors in pre-game basketball outcome models.

4.4 RQ3: Explainability Integration for Pre-Game Predictions

To support transparency and human-centered interpretation, explainability is integrated directly into the pre-game prediction workflow. For each prediction, SHAP-based local feature attributions are computed to identify which inputs most influenced the model’s output.

To make these explanations accessible to non-technical users, the SHAP attributions are converted into short natural-language rationales by passing the explanation outputs to the OpenAI API. This process generates a concise, game-specific explanation describing why the model favors one team over the other, rather than presenting only a numerical win probability.

This approach addresses gaps identified in the PRISMA review, where many studies report feature importance but do not surface explanations in a user-facing or interpretable format. By pairing probabilistic predictions with natural-language explanations, the system supports clearer reasoning and improved trust in pre-game assessments.

5 Discussion

This study set out to examine how NBA win-prediction models can evolve from static, pre-game forecasts into adaptive and interpretable systems that update as games unfold. Rather than optimizing for predictive accuracy in isolation, the primary contribution lies in understanding how different machine-learning approaches behave across game phases and how their outputs can be communicated transparently to human users. Findings from both the PRISMA-guided literature synthesis and the empirical evaluation of the proof-of-concept system yield several insights relevant to real-time sports analytics.

5.1 Interpretation of Findings

Results from the empirical evaluation indicate that model suitability is strongly phase-dependent. For pre-game prediction, an RBF-kernel Support Vector Machine (SVM) achieved strong and stable performance, reaching validation accuracies of 79% in the 2024–2025 season setting and 81% in the current pre-game pipeline. These results demonstrate that relatively simple, well-regularized models can perform competitively when paired with carefully constructed feature representations and appropriate temporal splits. The SVM’s computational efficiency and stability further support its role as a reliable baseline for early-stage forecasts, where available information is limited to season averages, recent form, and pre-game ratings.

In contrast, ensemble-based approaches such as Random Forest, XGBoost, and LightGBM consistently emerge in the literature as strong performers in in-game and momentum-aware contexts. These models are particularly effective at capturing non-linear interactions among performance indicators such as shooting efficiency, rebounding, turnovers, and score differential, which become increasingly informative as a game progresses. Importantly, the findings reinforce that no single algorithm dominates across all phases of a game. Instead, phase-aware modeling—where simpler models anchor pre-game predictions and more expressive ensemble methods are introduced as live data accumulates—offers a more robust and interpretable strategy than relying on a single, static predictor throughout an entire contest.

5.2 Feature Integration and Adaptability

The effectiveness of the selected pre-game SVM is closely tied to feature design. Results show that constructing features as home-team minus away-team differentials, combined with rolling ten-game recent-form statistics and a pre-game Elo rating, provides a compact yet expressive representation of team strength and momentum. These features enable the model to capture both long-term quality and short-term trends without introducing excessive noise.

More broadly, evidence from both the literature synthesis and system design confirms that incorporating in-game performance metrics substantially improves adaptability relative to models based solely on season-level averages. Features such as score differential, shooting percentages, rebounds, and assists provide immediate signals about game flow that static statistics cannot capture. Crucially, the relative importance of these features shifts over time. Early in games, efficiency and assist-related indicators tend to dominate, while rebounding, turnovers, and margin control become more influential in later phases. This nonstationary behavior underscores the limitations of static models and motivates adaptive systems that update both predictions and explanations as the game unfolds.

5.3 Explainability and Human Trust

A central contribution of this work is the integration of explainability directly into the prediction pipeline. For the pre-game model, SHAP-based local explanations are computed for each prediction and translated into short natural-language rationales. Rather than presenting users with a single win probability, the system highlights which features most influenced the prediction for a given matchup, grounding model output in recognizable basketball concepts such as shooting efficiency, recent performance, and team strength.

These findings reinforce the idea that explainability serves as a bridge between model output and human trust. However, the results also align with prior human-centered AI research showing that transparency alone does not guarantee calibrated trust. Explanations must be aligned with user expectations and domain knowledge to be effective. By focusing on interpretable, basketball-relevant features and presenting explanations at the individual game level, the framework supports more meaningful human–AI collaboration, enabling users to understand not only what the model predicts, but why that prediction is produced.

5.4 Limitations

Several limitations should be acknowledged. First, while explainability is implemented in the pre-game pipeline, this study does not include a controlled user study evaluating how explanations influence user understanding, trust calibration, or decision-making. As a result, claims regarding human trust are grounded in established literature rather than direct behavioral evidence.

Second, comparisons across algorithms are constrained by heterogeneity in datasets, feature sets, and evaluation protocols both in prior work and in the current system design. Although the PRISMA-guided review identifies consistent trends, it does not support a formal meta-analysis or definitive ranking of models across all settings. Additionally, real-time prediction systems remain sensitive to data latency and API reliability, which may affect live deployment performance.

Finally, the current implementation evaluates predictions at discrete update intervals rather than continuously at the possession or tracking-data level. While this design balances interpretability and computational efficiency, it may miss finer-grained dynamics observable in play-by-play or spatiotemporal data.

5.5 Future Work

Several directions for future work emerge from these findings. A primary extension is the deeper integration of XGBoost as the dominant in-game prediction model, leveraging its ability to capture non-linear interactions and rapidly adapt to momentum shifts. Future iterations will explore tighter coupling between the pre-game SVM and in-game XGBoost predictions, allowing the pre-game estimate to act as a structured prior that evolves throughout the game. Recent work in other sports has also formalized momentum as a measurable, time-evolving

signal, suggesting useful design patterns for cross-sport adaptation of stage-aware prediction pipelines [30].

Beyond basketball, the proposed framework is intentionally generalizable. Future work will extend the system to other sports with strong temporal structure, including soccer and American football, where phase transitions, momentum, and game state play a critical role in outcomes. Expanding coverage to playoff games and other high-stakes contexts is also a priority, as these settings introduce strategic and psychological dynamics that differ from regular-season play.

Additional extensions include incorporating richer spatiotemporal data, conducting user studies to evaluate explanation effectiveness, and refining visualization techniques to better support real-time sense-making. Together, these efforts aim to move beyond isolated prediction tasks toward scalable, multi-sport, explainable systems that support transparent and adaptive sports analytics.

6 Conclusion

This paper investigated how NBA win prediction can be framed as a phase-dependent task that benefits from both adaptive modeling and human-centered interpretability. Addressing **RQ1**, our PRISMA-guided synthesis indicates that no single algorithm consistently dominates across pre-game and in-game settings: classical models such as SVM remain competitive and stable baselines for pre-game forecasting, while ensemble methods (e.g., Random Forest, XGBoost, LightGBM) more often excel when live, nonlinear interactions become informative. In our implemented pre-game pipeline, an RBF-kernel SVM achieved **79%–81%** validation accuracy across two evaluation settings, supporting its use as a robust pre-game prior for the broader system.

Regarding **RQ2** and **RQ3**, the reviewed literature consistently shows that incorporating temporal and “recent form” signals improves adaptability beyond static season averages, and that interpretability tools such as SHAP help translate model behavior into domain-relevant reasons that users can understand. Building on these findings, we integrated SHAP-based local explanations into our pre-game workflow and generated concise natural-language rationales to support transparent, game-specific decision support. Overall, our contribution is a unified evidence-to-system framework that connects PRISMA-based synthesis to an explainable, phase-aware prediction pipeline—supporting practical, interpretable sports analytics without claiming to eliminate the inherent uncertainty that makes competitive sport compelling.

References

1. Zhao, K., Du, C., Tan, G.: Enhancing basketball game outcome prediction through fused graph convolutional networks and random forest algorithm. *Entropy* **25**(5), 765 (2023)
2. Ouyang, Y., Li, X., Zhou, W., Hong, W., Zheng, W., Qi, F., Peng, L.: Integration of machine learning XGBoost and SHAP models for NBA game outcome prediction and quantitative analysis methodology. *PLOS ONE* **19**(7), e0307478 (2024)

3. Hodge, V.J., Devlin, S., Sephton, N., Block, F., Cowling, P.I., Drachen, A.: Win prediction in multiplayer esports: Live professional match prediction. *IEEE Transactions on Games* **13**(4), 368–379 (2019)
4. Horvat, T., Job, J., Logoazar, R., Livada, Č.: A data-driven machine learning algorithm for predicting the outcomes of NBA games. *Symmetry* **15**(4), 798 (2023)
5. Thabtah, F., Zhang, L., Abdelhamid, N.: NBA game result prediction using feature analysis and machine learning. *Annals of Data Science* **6**(1), 103–116 (2019)
6. Wang, J.: Predictive analysis of NBA game outcomes through machine learning. In: *Proceedings of the 6th International Conference on Machine Learning and Machine Intelligence*, pp. 46–55 (2023)
7. Jain, S., Kaur, H.: Machine learning approaches to predict basketball game outcome. In: *Proceedings of the 3rd International Conference on Advances in Computing, Communication & Automation (ICACCA)*, pp. 1–7. IEEE (2017)
8. Perricone, J., Shaw, I., Swiechowicz, W.: Predicting results for professional basketball using NBA API data. Stanford CS229 Technical Report (2016)
9. Li, C., Zhang, H., Zhang, Y., Shen, J., An, R.: Artificial intelligence and machine learning-based predictive models for basketball game outcomes: A systematic review. *PLOS ONE* **20**(6), e0326326 (2025)
10. Page, M.J., McKenzie, J.E., Bossuyt, P.M., et al.: The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **372** (2021)
11. Rodríguez, J.A.: “This game is in the fridge”: Predicting NBA game outcomes. CS229 Machine Learning Course Project, Stanford University (2019)
12. Bunker, R., Susnjak, T.: A review of machine learning methods for sport outcome prediction. *Applied Artificial Intelligence* **36**(1) (2022)
13. Alves, R., Barbosa, H.: Benchmarking machine learning models for NBA and WNBA game outcome prediction. *Journal of Sports Analytics* (2025)
14. He, Z., Choi, S.-B.: Stacked ensemble learning for NBA game outcome prediction. *Expert Systems with Applications* (2025)
15. Zhao, K., Du, C., Tan, G.: Basketball outcome prediction using graph attention networks with feature selection. *Neural Computing and Applications* (2024)
16. McFarlane, M.: Evaluating win probability models in NBA end-game decision making. *Journal of Sports Analytics* (2019)
17. Sandholtz, N., Bornn, L.: Markov decision processes for basketball shot selection. *Journal of Quantitative Analysis in Sports* (2020)
18. Vračar, M., Štrumbelj, E., Kononenko, I.: Modeling basketball game outcomes using play-by-play data. *Expert Systems* (2016)
19. Kvam, P.H., Sokol, J.S.: A logistic regression/Markov chain model for NCAA basketball outcomes. *Naval Research Logistics* (2006)
20. Štrumbelj, E., Kononenko, I.: A dynamic model for estimating win probabilities in basketball. *Journal of Quantitative Analysis in Sports* (2012)
21. Jiao, Y., et al.: Using SportVU tracking data to predict basketball possession outcomes. *IEEE Transactions on Knowledge and Data Engineering* (2025)
22. Sano, S., et al.: Predicting pass receivability in basketball using player movement. In: *Proceedings of the MIT Sloan Sports Analytics Conference* (2019)
23. García-Méndez, J., de Arriba-Pérez, F.: Explainable streaming win prediction with sliding windows. *Information Visualization* (2025)
24. Du, F., Yuan, X.: A survey of visual analytics in sports. *IEEE Computer Graphics and Applications* (2020)
25. Kim, B., et al.: Human-centered evaluation of explainable AI systems: A systematic review. *ACM Computing Surveys* (2024)

26. Kranzinger, S., et al.: Explainable artificial intelligence in sports science: A scoping review. *Sports Medicine* (2025)
27. Hammes, D., et al.: Artificial intelligence in elite sports: Applications and challenges. *Sports Engineering* (2022)
28. van Haaren, J.: Why trust your numbers? Explainable AI in sports analytics. *AI Magazine* (2021)
29. Bunker, R., Susnjak, T.: The application of machine learning techniques for predicting match results in team sport: A review. *Journal of Artificial Intelligence Research* **73**, 1285–1322 (2022)
30. Long, Z., Li, N., Sun, M., Luo, J., Pan, T., Yin, Y., Yang, X.: A tennis momentum analysis method based on Gaussian dynamics and machine learning. In: *Proceedings of the 3rd International Conference on Algorithms, Data Mining, and Information Technology*, pp. 13–17 (2024)